PROJECT DOCUMENTATION

STRATEGIC PRODUCT PLACEMENT ANALYSIS

Retail Analytics • Tableau • Flask Web Integration

Project Overview

Strategic Product Placement Analysis is an interactive retail analytics project developed to study how product category, store position, consumer demographics, foot traffic, pricing, promotions, and seasonal factors relate to sales performance. The project transforms retail data into Tableau visualizations, an interactive dashboard, and a guided analytical story. The published Tableau views are integrated into a Flask and Bootstrap web application for clear project presentation and interactive exploration.

Project Objective

The objective is to simplify the interpretation of retail product placement data and present meaningful sales patterns through interactive visual analysis. The solution supports comparison of retail factors and provides a structured view of product placement performance.

Group Project Development

This project was completed as an group implementation.

Although multiple members were assigned to the group, all team members actively collaborated and contributed throughout the project development period. As a team, we jointly designed, developed, tested, documented, deployed, and organized the entire project. The Tableau dashboard, Tableau Story, Flask web integration, project documentation, testing, and deployment were successfully completed as part of the assigned project work.

Problem Statement

Retail product performance is affected by several connected factors. When product placement, customer, traffic, price, promotion, and seasonal information is examined only as raw data or disconnected reports, identifying useful patterns becomes difficult. A visual and interactive analytical solution is required to compare these factors and communicate important findings clearly.

Proposed Solution

The proposed solution uses Tableau to analyze the retail dataset and create business-question visualizations. These visualizations are combined into an interactive dashboard and a three-view analytical story. Tableau Public hosts the published views, while Python and Flask serve a Bootstrap-based website that embeds the dashboard and story.

PROJECT DEVELOPMENT

Development Process and Major Components

Phase	Work Completed
Ideation	Defined the retail analytics problem, user needs, and possible product placement analysis ideas.
Requirement Analysis	Identified functional, data, visualization, web integration, and system requirements.
Project Design	Prepared the problem–solution fit, proposed solution, data flow, and layered solution architecture.
Data Analysis	Reviewed retail fields and defined business questions for category, position, demographics, traffic, price, promotion, and season.
Tableau Development	Created visualizations, dashboard filters, the final interactive dashboard, and three Tableau Story views.
Web Development	Customized a Bootstrap interface and used Flask to render the project website.
Integration	Embedded the Tableau dashboard and Tableau Story into the Flask web application.
Deployment	Published the Tableau views and deployed the Flask application as a public Render web service.
Testing	Checked website loading, navigation, dashboard interaction, Story navigation, embedded views, and public access.
Documentation	Prepared project phase documents, screenshots, links, testing records, and final project documentation.

Technology Stack

Technology	Role in the Project
Tableau Public	Retail analytics, charts, dashboard, filters, Story development, and visualization publishing.
Python	Runtime environment for the Flask web application.
Flask	Web framework used to serve and render the project interface.
HTML and CSS	Website structure and visual customization.
Bootstrap	Responsive layout, navigation, and frontend components.
JavaScript	Frontend interaction and embedded visualization loading support.
VS Code	Development, project organization, editing, and terminal execution.
GitHub	Source code and project repository management.

Technology	Role in the Project
Render	Public deployment of the Flask web application.

ANALYSIS AND RESULTS

Business Questions and Visualization Outcomes

Analysis Area	Project Analysis
Product Category	Compared average sales volume across Clothing, Electronics, and Food.
Product Position	Analyzed placement performance across Aisle, End-cap, and Front of Store.
Consumer Demographics	Compared sales performance across customer demographic groups.
Foot Traffic	Studied sales patterns across High, Medium, and Low foot-traffic levels.
Pricing	Compared product price with competitor price and reviewed category-level pricing.
Promotion	Reviewed sales performance in relation to promotional activity.
Season	Compared sales performance across seasonal categories.
Combined Placement Patterns	Used the dashboard and Story views to present connected product placement insights.

Final Deliverables

• Interactive Tableau Dashboard with filters and retail performance visualizations.

• Three-view Tableau Story covering category performance, foot traffic and position, and product placement distribution.

• Flask and Bootstrap project portfolio with embedded Tableau views.

• Public Tableau Dashboard and Tableau Story links.

• Public Render deployment of the project web application.

• Retail dataset used for the analysis.

• Phase-wise project documentation, testing template, screenshots, and project links document.

Testing Result

The final project was tested for public website loading, section navigation, Tableau Dashboard display and interaction, Tableau Story loading and scene navigation, embedded visualization alignment, page refresh, and public deployment access. The defined normal-use project test cases were recorded as PASS.

Project Outcome

The final project provides a complete interactive retail analytics experience.

Retail data is transformed into Tableau visual insights, the dashboard supports interactive exploration, and the Story presents important findings in a guided sequence. Flask and Bootstrap provide the presentation interface, while public deployment makes the project accessible for demonstration. The complete solution, including planning, analytics, visualization, integration, testing, deployment, and documentation, was developed independently as individual project work.

Conclusion

Strategic Product Placement Analysis demonstrates the use of data visualization and web integration to communicate retail sales and product placement patterns. The project combines Tableau analytics with a Flask-based portfolio and presents the work through a structured development and documentation process. The completed implementation is suitable for project demonstration and submission.

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